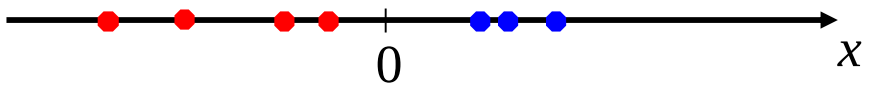
Support Vector Machine

Contents

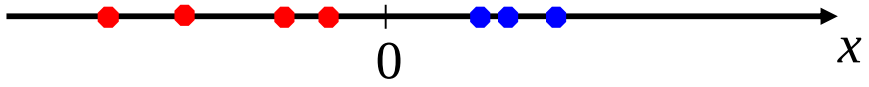
- Linear Support Vector Machine
- Learning Linear SVM
- Dual Form for Learning LSVM
- Linear SVM with Soft Margin
- Non-Linear SVM
- Kernel Trick
- Some more on Kernels
- Discussion

Non-Linear SVM

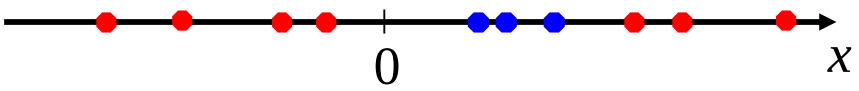
- **Data that are linearly separable**



- **Data with noise**
  - linearly separable considering error



- **What about this?**



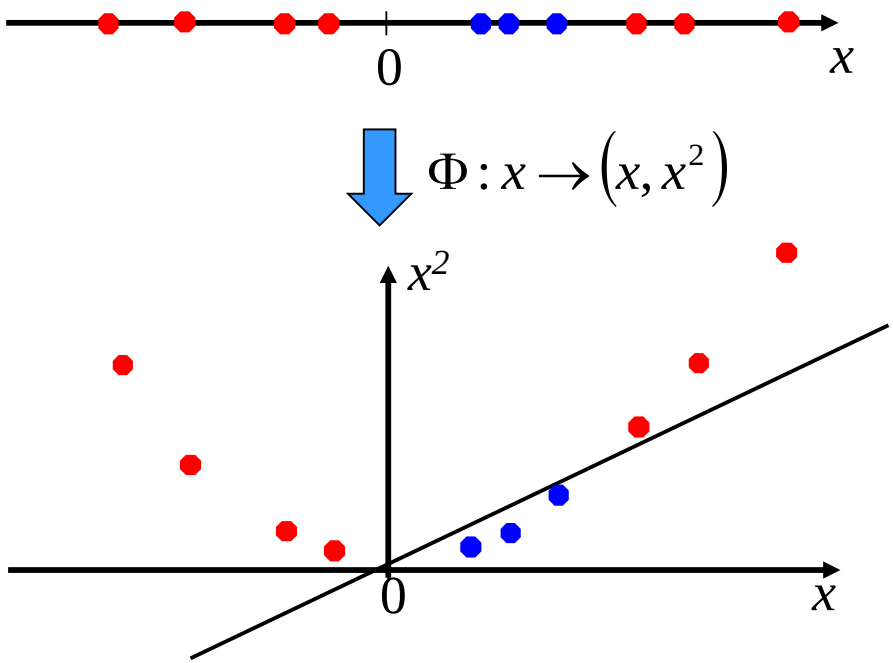
We need a non-linear boundary!!

But, How??

Non-Linear SVM

- **Trick**

1. Map data to a higher-dimensional space

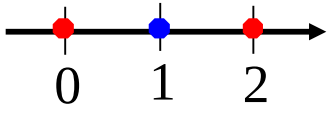


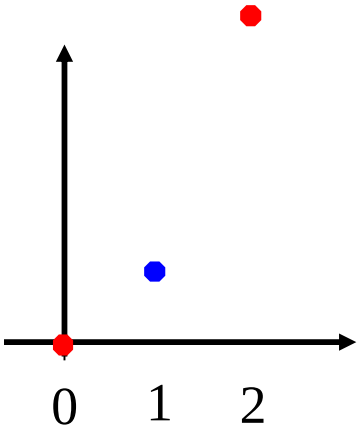
Can we find such mapping easily?

2. Find a linear boundary in the higher-dimensional space

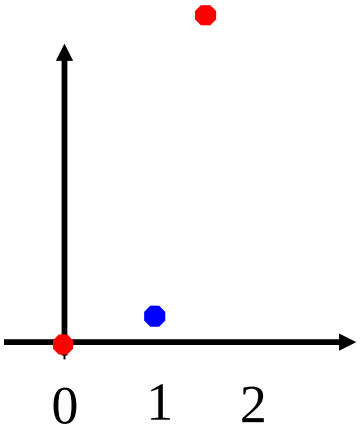
Non-Linear SVM

- **Most of non-linear mapping do this!!**

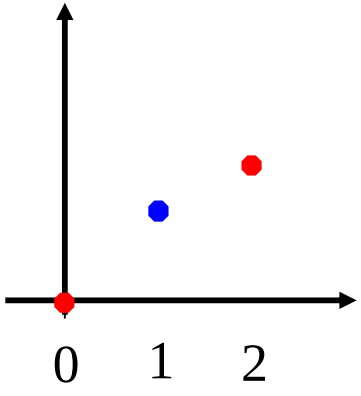




$\Phi : x \to \left(x, x^2\right)$



$\Phi : x \to \left(\sqrt{x}, e^x\right)$

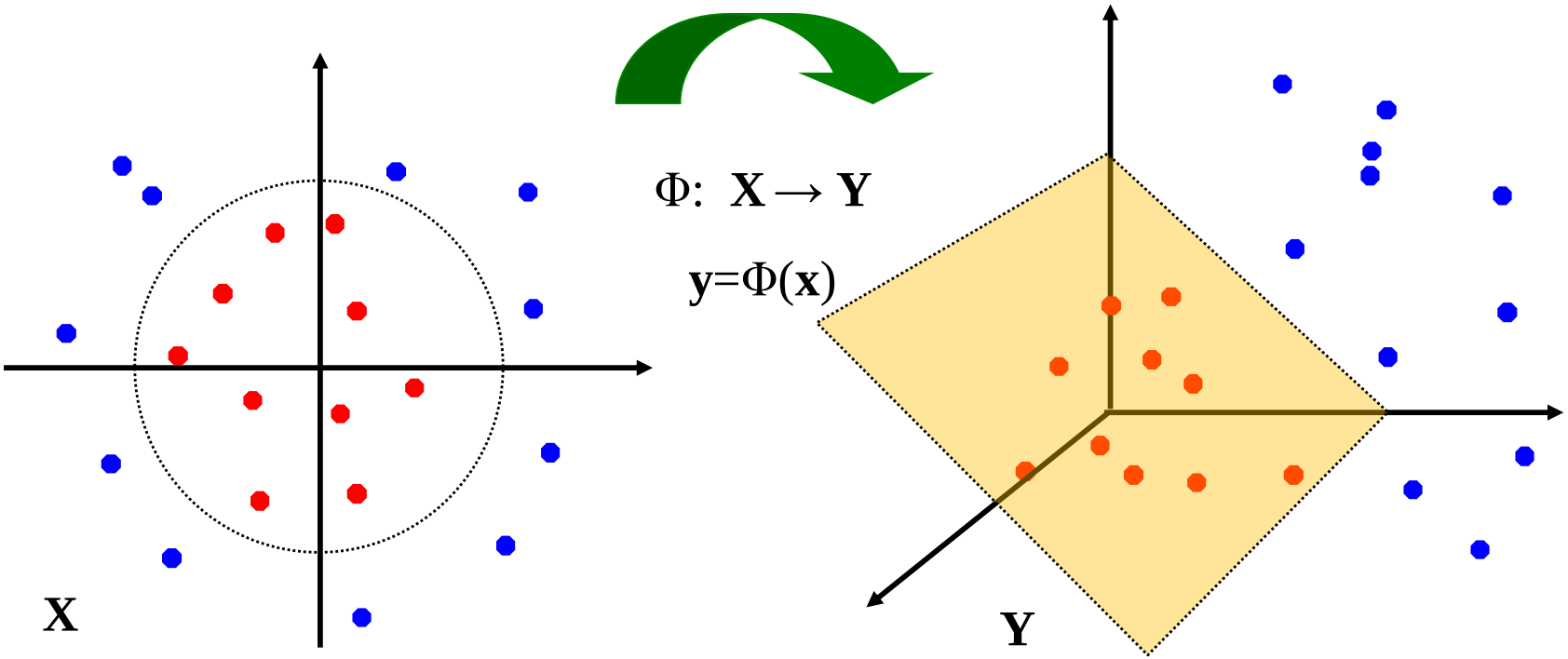


$\Phi : x \to \left(x, \sqrt{x}\right)$

Then, How high dimension?
- The higher dimension, the better

Non-Linear SVM

- **General idea**
  - the original input space can always be mapped to some higher-dimensional feature space where the training set is separable:



Non-Linear SVM

- **Remind: Formulation for linear boundary**
  - For given $D = \{(\mathbf{x}_1, y_1), \cdots, (\mathbf{x}_n, y_n)\}$ where $y_i \in \{-1, +1\}$

$$\max_{\alpha_1, \cdots, \alpha_n} \left(\sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j (\mathbf{x}_i \cdot \mathbf{x}_j) \right)$$

$$\text{subject to} \begin{cases} \sum_{i=1}^{n} \alpha_i y_i = 0 \\ 0 \le \alpha_i \le C \quad i = 1, \cdots, n \end{cases}$$

  - Solution

$$\mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i$$

$$b = y_k - \mathbf{w} \cdot \mathbf{x}_k \quad \text{for any } \mathbf{x}_k \text{ such that } 0 < \alpha_k < C$$

Non-Linear SVM

- **Formulation for non-linear boundary**
  1. For given $D = \{(\mathbf{x}_1, y_1), \cdots, (\mathbf{x}_n, y_n)\}$ where $y_i \in \{-1, +1\}$
  2. Define Φ for $\Phi : \mathbf{x} \rightarrow \Phi(\mathbf{x})$
  3. Convert data using Φ

$$D = \{(\Phi(\mathbf{x}_1), y_1), \cdots, (\Phi(\mathbf{x}_n), y_n)\} \quad \text{where } y_i \in \{-1, +1\}$$

  4. Solve

$$\arg\max_{\alpha_1, \cdots, \alpha_n} \left(\sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \left(\Phi(\mathbf{x}_i) \cdot \Phi(\mathbf{x}_j) \right) \right)$$

$$\text{subject to } \begin{cases} \sum_{i=1}^{n} \alpha_i y_i = 0 \\ 0 \le \alpha_i \le C \quad i = 1, \cdots, n \end{cases}$$

Non-Linear SVM

- **Formulation for non-linear boundary**

  5. Solution for the boundary

$$\mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i \Phi(\mathbf{x}_i)$$

$$b = y_k - \mathbf{w} \cdot \Phi(\mathbf{x}_k)$$

$$= y_k - \sum_{i=1}^{n} \alpha_i y_i \Phi(\mathbf{x}_i) \cdot \Phi(\mathbf{x}_k) \text{ for any } \Phi(\mathbf{x}_k) \text{ such that } 0 < \alpha_k < C$$

Non-Linear SVM

- **Example**

$$D = \{(1,1,-1),(2,2,-1),(3,1,-1),(1,2,1),(3,2,1)\}$$

$$\Phi : (x_1, x_2) \rightarrow (x_1, x_2, x_1^2, x_2^2, x_1^3, x_2^3, x_1x_2, x_1x_2^2, x_1^2x_2)$$

$$\Phi(D) = \left\{ \begin{array}{l} (1,1,1,1,1,1,1,1,1,-1),(2,2,4,4,8,8,4,8,8,-1), \\ (3,1,9,1,27,1,3,2,9,-1),(1,2,1,4,1,8,2,4,2,1), \\ (3,2,9,4,27,8,6,12,18,1) \end{array} \right\}$$

$$\arg\max_{\alpha_1,\cdots,\alpha_n} \left(\sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \left(\Phi(\mathbf{x}_i) \cdot \Phi(\mathbf{x}_j) \right) \right)$$

$$\text{subject to} \left\{ \begin{array}{l} \sum_{i=1}^{n} \alpha_i y_i = 0 \\ 0 \le \alpha_i \le C \qquad i = 1, \cdots, n \end{array} \right.$$

Non-Linear SVM

- **Observation: Any severe difference??**
  - Original Formulation

$$D = \{(1,1,-1),(2,2,-1),(3,1,-1),(1,2,1),(3,2,1)\}$$

$$\arg\max_{\alpha_1,\cdots,\alpha_n}\left(\sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j(\mathbf{x}_i\cdot\mathbf{x}_j)\right) \quad \text{,subject to } \begin{cases}\sum_{i=1}^{n}\alpha_i y_i = 0 \\ 0 \le \alpha_i \le C \qquad i = 1,\cdots,n\end{cases}$$

  - After mapping into a higher space

$$\Phi(D) = \begin{Bmatrix}(1,1,1,1,1,1,1,1,1,-1),(2,2,4,4,8,8,4,8,8,-1),(3,1,9,1,27,1,3,2,9,-1),\\ (1,2,1,4,1,8,2,4,2,1),(3,2,9,4,27,8,6,12,18,1)\end{Bmatrix}$$

$$\arg\max_{\alpha_1,\cdots,\alpha_n}\left(\sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j(\Phi(\mathbf{x}_i)\cdot\Phi(\mathbf{x}_j))\right) \quad \text{,subject to } \begin{cases}\sum_{i=1}^{n}\alpha_i y_i = 0 \\ 0 \le \alpha_i \le C \qquad i = 1,\cdots,n\end{cases}$$

Non-Linear SVM

- **Observation: Which one do we need?**
  - Mapping data into a higher space

$$\Phi : \mathbf{x} \rightarrow \Phi(\mathbf{x})$$

  - Inner products of mapped data

$$\Phi(\mathbf{x}_i) \cdot \Phi(\mathbf{x}_j)$$

$$\Phi(D) = \left\{ \begin{array}{l} (1,1,1,1,1,1,1,1,1,-1), (2,2,4,4,8,8,4,8,8,-1), (3,1,9,1,27,1,3,2,9,-1), \\ (1,2,1,4,1,8,2,4,2,1), (3,2,9,4,27,8,6,12,18,1) \end{array} \right\}$$

$$\arg\max_{\alpha_1,\cdots,\alpha_n} \left(\sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \left(\Phi(\mathbf{x}_i) \cdot \Phi(\mathbf{x}_j) \right) \right) \text{, subject to } \left\{ \begin{array}{l} \sum_{i=1}^{n} \alpha_i y_i = 0 \\ 0 \le \alpha_i \le C \quad i = 1, \cdots, n \end{array} \right.$$

Kernel Trick

- **Another Transform**
  - Instead of

$$\Phi : (x_1, x_2) \rightarrow (x_1, x_2, x_1^2, x_2^2, x_1^3, x_2^3, x_1 x_2, x_1 x_2^2, x_1^2 x_2)$$

  - What about this?

$$\Phi : (x_1, x_2) \rightarrow (1, \sqrt{3}x_1, \sqrt{3}x_2, \sqrt{3}x_1^2, \sqrt{3}x_2^2, x_1^3, x_2^3, \sqrt{6}x_1 x_2, \sqrt{3}x_1 x_2^2, \sqrt{3}x_1^2 x_2)$$

    - Why??
    - Evaluate $\Phi(\mathbf{x}_i) \cdot \Phi(\mathbf{x}_j)$

Kernel Trick

- **Another Transform**

$$\mathbf{x}_1 = (x_{11}, x_{12}),\ \mathbf{x}_2 = (x_{21}, x_{22})$$

$$\Phi(\mathbf{x}_1) = (1, \sqrt{3}x_{11}, \sqrt{3}x_{12}, \sqrt{3}x_{11}^2, \sqrt{3}x_{12}^2, x_{11}^3, x_{12}^3, \sqrt{6}x_{11}x_{12}, \sqrt{3}x_{11}x_{12}^2, \sqrt{3}x_{11}^2x_{12})$$

$$\Phi(\mathbf{x}_2) = (1, \sqrt{3}x_{21}, \sqrt{3}x_{22}, \sqrt{3}x_{21}^2, \sqrt{3}x_{22}^2, x_{21}^3, x_{22}^3, \sqrt{6}x_{21}x_{22}, \sqrt{3}x_{21}x_{22}^2, \sqrt{3}x_{21}^2x_{22})$$

$$\begin{aligned}\Phi(\mathbf{x}_2)\cdot\Phi(\mathbf{x}_2) &= 1 + 3x_{11}x_{21} + 3x_{12}x_{22} + 3x_{11}^2x_{21}^2 + 3x_{12}^2x_{22}^2 + x_{11}^3x_{21}^3 + x_{12}^3x_{22}^3 \\ &\quad + 6x_{11}x_{12}x_{21}x_{22} + 3x_{11}x_{12}^2x_{21}x_{22}^2 + 3x_{11}^2x_{12}x_{21}^2x_{22} \\ &= (x_{11}x_{21} + x_{12}x_{22})^3 + 3(x_{11}x_{21} + x_{12}x_{22})^2 + 3(x_{11}x_{21} + x_{12}x_{22})^2 + 1 \\ &= ((x_{11}x_{21} + x_{12}x_{22}) + 1)^3 \\ &= (\mathbf{x}_1 \cdot \mathbf{x}_2 + 1)^3\end{aligned}$$

What does it mean??

If a transform function is well designed, we can easily evaluate the inner products!!

Kernel Trick

- **Example**

$$\Phi(\mathbf{x}_1) = (1,\ \sqrt{3},\ \sqrt{3},\ \sqrt{3},\ \sqrt{3},\ 1,\ 1,\ \sqrt{6},\ \sqrt{3},\ \sqrt{3})$$
$$\Phi(\mathbf{x}_2) = (1,\ 2\sqrt{3},\ 2\sqrt{3},\ 4\sqrt{3},\ 4\sqrt{3},\ 8,\ 8,\ 4\sqrt{6},\ 8\sqrt{3},\ 8\sqrt{3})$$

$$\mathbf{x}_1 = (1,\ 1)$$
$$\mathbf{x}_2 = (2,\ 2)$$

$$\Phi(\mathbf{x}_2)\cdot\Phi(\mathbf{x}_2) = 1+6+6+12+12+8+8+24+24+24 = 125$$

$$\| \quad \|$$

$$K(\mathbf{x}_1,\mathbf{x}_2) = (\mathbf{x}_1\cdot\mathbf{x}_2+1)^3 = (4+1)^3 = 125$$

Kernel Trick

- **Another Transform**
  - Instead of solving this

$$D = \{(x_{11}, x_{12}, y_1), \cdots, (x_{n1}, x_{n2}, , y_n)\} \quad \text{where } y_i \in \{-1, +1\}$$

$$\Phi : (x_1, x_2) \rightarrow (1, \sqrt{3}x_1, \sqrt{3}x_2, \sqrt{3}x_1^2, \sqrt{3}x_2^2, x_1^3, x_2^3, \sqrt{6}x_1x_2, \sqrt{3}x_1x_2^2, \sqrt{3}x_1^2x_2)$$

$$\arg\max_{\alpha_1, \cdots, \alpha_n} \left(\sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \left(\Phi(\mathbf{x}_i) \cdot \Phi(\mathbf{x}_j) \right) \right) \text{, subject to } \begin{cases} \sum_{i=1}^{n} \alpha_i y_i = 0 \\ 0 \le \alpha_i \le C \end{cases} \quad i = 1, \cdots, n$$

  - Solve this. It saves your times!!

$$D = \{(x_{11}, x_{12}, y_1), \cdots, (x_{n1}, x_{n2}, y_n)\} \quad \text{where } y_i \in \{-1, +1\}$$

$$\arg\max_{\alpha_1, \cdots, \alpha_n} \left(\sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \left(\mathbf{x}_i \cdot \mathbf{x}_j + 1 \right)^3 \right) \text{, subject to } \begin{cases} \sum_{i=1}^{n} \alpha_i y_i = 0 \\ 0 \le \alpha_i \le C \end{cases} \quad i = 1, \cdots, n$$

Kernel Trick

- **Another Transform**

Kernel	$\Phi(\mathbf{x}_i)$ when $\mathbf{x}=(x_{i1}, x_{i2})$
$(\mathbf{x}_i \cdot \mathbf{x}_j+1)^2$	$\left(1, \sqrt{2}x_{i1}, \sqrt{2}x_{i2}, x_{i1}^2, x_{i2}^2, \sqrt{2}x_{i1}x_{i2}\right)$
$(\mathbf{x}_i \cdot \mathbf{x}_j+1)^3$	$\left(1, \sqrt{3}x_{i1}, \sqrt{3}x_{i2}, \sqrt{3}x_{i1}^2, \sqrt{3}x_{i2}^2, \sqrt{6}x_{i1}x_{i2}, x_{i1}^3, x_{i2}^3, \sqrt{3}x_{i1}x_{i2}^2, \sqrt{3}x_{i1}^2x_{i2}\right)$
$(\mathbf{x}_i \cdot \mathbf{x}_j+1)^4$	$\left(1, 2x_{i1}, 2x_{i2}, \sqrt{6}x_{i1}^2, \sqrt{6}x_{i2}^2, \sqrt{12}x_{i1}x_{i2}, 2x_{i1}^3, 2x_{i2}^3, \sqrt{12}x_{i1}x_{i2}^2, \sqrt{12}x_{i1}^2x_{i2}, \cdots\right)$

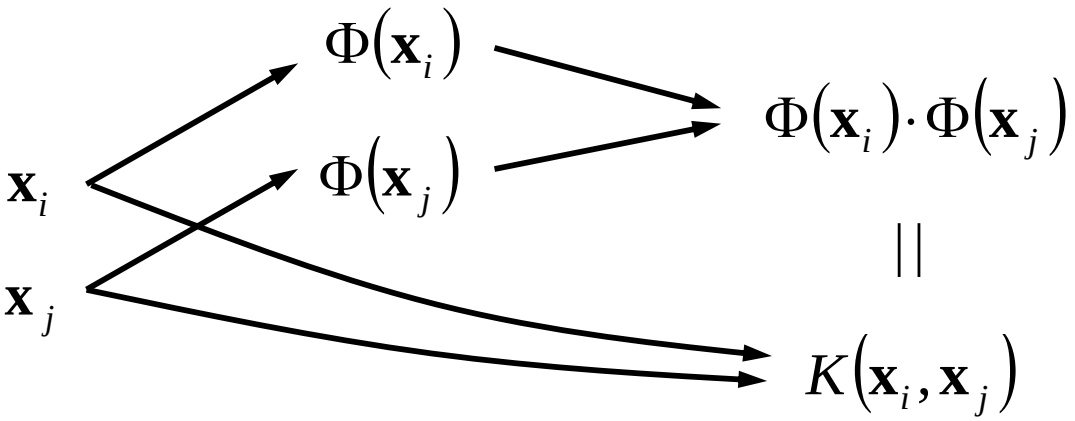
Do we really need to know the transform function?

- If $\mathbf{x}_i$ and $\mathbf{x}_j$ are m-dimensional and I use $(\mathbf{x}_i \cdot \mathbf{x}_j+1)^k$

$$\Phi(\mathbf{x}) \text{ is } \frac{(m+k)!}{m!k!} \text{ dimensiona l}$$

Kernel Trick

- **Kernel**
  - A function that corresponds to a dot product of two feature vectors in some expanded feature space
  - You have two functions $\Phi(\mathbf{x})$ and $K(\mathbf{x},\mathbf{y})$ and it happens that $\Phi(\mathbf{x})\cdot\Phi(\mathbf{y})=K(\mathbf{x},\mathbf{y})$ Then, *K* is called a kernel function



Kernel Trick

- **Final Formulation of SVM**
  - For given $D = \{(\mathbf{x}_1, y_1), \cdots, (\mathbf{x}_1, y_n)\}$ where $y_i \in \{-1, +1\}$
    Choose *C* and *K*, and Solve

$$\arg\max_{\alpha_1, \cdots, \alpha_n} \left(\sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \right)$$

$$\text{subject to} \begin{cases} \sum_{i=1}^{n} \alpha_i y_i = 0 \\ 0 \le \alpha_i \le C \qquad i = 1, \cdots, n \end{cases}$$

  - Solution for the boundary

$$\mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i \Phi(\mathbf{x}_i)$$

$$b = y_k - \mathbf{w} \cdot \Phi(\mathbf{x}_k)$$

$$= y_k - \sum_{i=1}^{n} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}_j) \text{ for any } k \text{ such that } 0 < \alpha_k < C$$

For unknown *x*

$$\text{Class of } \mathbf{x} = \begin{cases} +1 & \text{if } y(\mathbf{x}) \ge 0 \\ -1 & \text{if } y(\mathbf{x}) < 0 \end{cases}$$

$$\text{where } y(\mathbf{x}) = \sum_{i=1}^{n} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$$

Some More on Kernels

- **Examples of commonly-used kernel functions:**
  - Linear Kernel

$$K(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i \cdot \mathbf{x}_j$$

  - Homogeneous Polynomial Kernel

$$K(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i \cdot \mathbf{x}_j)^d$$

  - Polynomial Kernel

$$K(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i \cdot \mathbf{x}_j)^d$$

  - Gaussian Kernel (RBF Kernel)

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$$

  - Sigmoid Kernel

$$K(\mathbf{x}_i, \mathbf{x}_j) = \tanh(a\,\mathbf{x}_i \cdot \mathbf{x}_j + b)$$

Some More on Kernels

- **Comments on Some Kernels**
  - Homogeneous Polynomial Kernel

$$K(\mathbf{x},\mathbf{y}) = (\mathbf{x}\cdot\mathbf{y})^d \qquad \Phi(\mathbf{x}) = \left(x_1^2, \sqrt{2}x_1x_2, x_2^2\right) \quad \Phi(\mathbf{y}) = \left(y_1^2, \sqrt{2}y_1y_2, y_2^2\right)$$

  - RBF Kernel

$$K(\mathbf{x},\mathbf{y}) = \exp\left(-\frac{\|\mathbf{x}-\mathbf{y}\|^2}{2}\right) = \exp\left(-\frac{(x_1-y_1)^2+(x_2-y_2)^2}{2}\right)$$

$$= \exp\left(-\frac{x_1^2-2x_1y_1+y_1^2+x_2^2-2x_2y_2+y_2^2}{2}\right)$$

$$= \exp\left(-\frac{\|x\|^2}{2}\right)\exp\left(-\frac{\|y\|^2}{2}\right)\exp(\mathbf{x}\cdot\mathbf{y})$$

$$= \exp\left(-\frac{\|x\|^2}{2}\right)\exp\left(-\frac{\|y\|^2}{2}\right)\sum_{n=0}^{\infty}\frac{(\mathbf{x}\cdot\mathbf{y})^n}{n!}$$

Some More on Kernels

- **Which functions can be Kernels?**
  - Could K(a,b) = $(a-b)^3$ be a kernel function ?

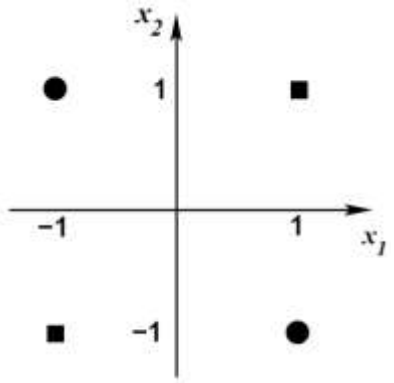
- **Mercer's Condition**
  - If a function, *K(x, y)*, is positive semi-definite function and, for any function *f(x)* whose $\int f^2(x)\,dx$ is finite, the following inequality holds

$$\int f(x)K(x,y)f(y)\,dxdy > 0$$

  - Then, *K(x,y)* is a kernel
    - That is, there is a Φ which satisfies $K(x,y)=\Phi(x)\cdot\Phi(y)$
    - What is Φ?

Some More on Kernels

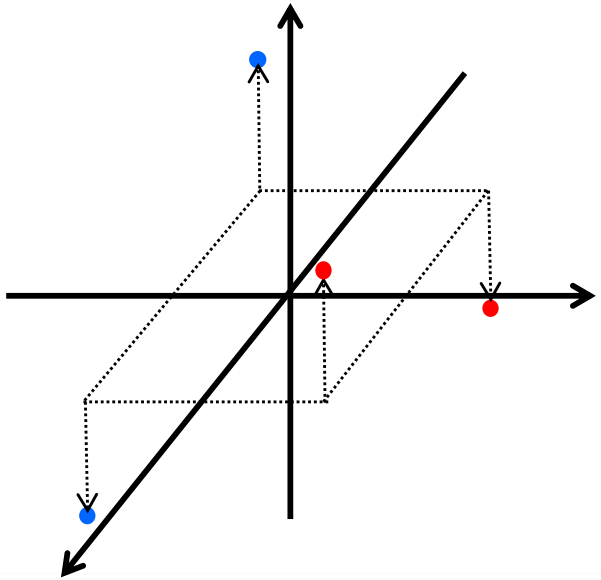
- **XOR Example**



$$K(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i \cdot \mathbf{x}_j)^d$$

$$\Phi : (x_1, x_2) \rightarrow (x_1^2, x_2^2, \sqrt{2}x_1x_2, \sqrt{2}x_1, \sqrt{2}x_2, 1)$$

x_i	y_i	$\Phi(x_i)$
(1,1)	-1	(1, 1,√2, √2, √2, 1)
(1,-1)	1	(1, 1,-√2, √2, –√2, 1)
(-1,1)	1	(1, 1,-√2, –√2, √2, 1)
(-1,-1)	-1	(1, 1,√2, –√2, –√2, 1)



Some More on Kernels

- **XOR Example: Dual SVM Formulization**

$$\arg\max_{\alpha_1,\cdots,\alpha_n}\left(\sum_{i=1}^{4}\alpha_i-\frac{1}{2}\sum_{i=1}^{4}\sum_{j=1}^{4}\alpha_i\alpha_j y_i y_j K(\mathbf{x}_i,\mathbf{x}_j)\right)$$

$$\arg\max_{\alpha_1,\cdots,\alpha_n}\left(\sum_{i=1}^{n}\alpha_i-\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j K(\mathbf{x}_i,\mathbf{x}_j)\right)$$

$$\text{subject to}\begin{cases}\sum_{i=1}^{n}\alpha_i y_i=0\\ 0\le\alpha_i \qquad i=1,\cdots,n\end{cases}$$

$$=\arg\max_{\alpha_1,\cdots,\alpha_n}\left(\alpha_1+\alpha_2+\alpha_3+\alpha_4-\frac{1}{2}\begin{pmatrix}9\alpha_1^2-2\alpha_1\alpha_2-2\alpha_1\alpha_3+2\alpha_1\alpha_4+\\ 9\alpha_2^2+2\alpha_2\alpha_3-2\alpha_2\alpha_4+\\ 9\alpha_3^2-2\alpha_3\alpha_4+\\ 9\alpha_4^2\end{pmatrix}\right)$$

$$\text{subject to}\begin{cases}-\alpha_1+\alpha_2+\alpha_3-\alpha_4=0\\ 0\le\alpha_i \qquad i=1,\cdots,n\end{cases}$$

$$F(\alpha_1,\alpha_2,\alpha_3,\alpha_4)$$

$$G(\alpha_1,\alpha_2,\alpha_3,\alpha_4)$$

Some More on Kernels

- **XOR Example: Solve** $\arg\max_{\alpha_1,\cdots,\alpha_n}(F-\lambda G)$

$$\frac{\partial L}{\partial \alpha_1}=0 \Leftrightarrow 9\alpha_1-\alpha_2-\alpha_3+\alpha_4-\lambda=1$$
$$\frac{\partial L}{\partial \alpha_2}=0 \Leftrightarrow \alpha_1-9\alpha_2-\alpha_3+\alpha_4+\lambda=-1$$
$$\frac{\partial L}{\partial \alpha_3}=0 \Leftrightarrow \alpha_1-\alpha_2-9\alpha_3+\alpha_4-\lambda=-1$$
$$\frac{\partial L}{\partial \alpha_4}=0 \Leftrightarrow \alpha_1-\alpha_2-\alpha_3+9\alpha_4+\lambda=1$$
$$\frac{\partial L}{\partial \lambda}=0 \Leftrightarrow -\alpha_1+\alpha_2+\alpha_3-\alpha_4=0$$

$$\alpha_1=\alpha_2=\alpha_3=\alpha_4=\frac{1}{8}$$

x_i	y_i	$\Phi(x_i)$
(1,1)	-1	(1, 1,√2, √2, √2, 1)
(1,-1)	1	(1, 1,-√2, √2, –√2, 1)
(-1,1)	1	(1, 1,-√2, –√2, √2, 1)
(-1,-1)	-1	(1, 1,√2, –√2, –√2, 1)

$$\mathbf{w}=\sum_{i=1}^{n}\alpha_i y_i \Phi(\mathbf{x}_i)$$
$$b=y_k-\mathbf{w}\cdot\Phi(\mathbf{x}_k)$$
$$=y_k-\sum_{i=1}^{n}\alpha_i y_i K(\mathbf{x}_i,\mathbf{x}_j) \text{ for any } k \text{ such that } 0<\alpha_k<C$$

Discussion

- **SVM with RBF Kernels vs. RBFNN**

Boundary of SVM

$$y = \mathbf{w}\cdot\mathbf{x} + b$$

$$\text{where } \mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i \Phi(\mathbf{x}_i)$$

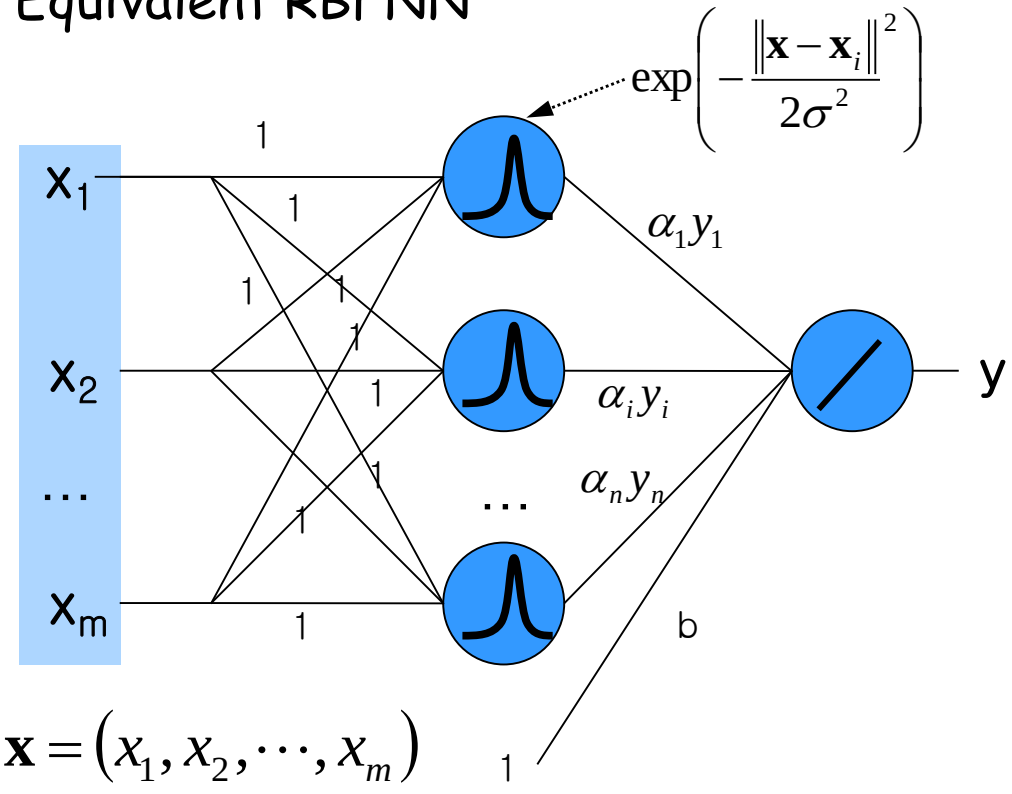
$$y = \Phi(\mathbf{x})\cdot\sum_{i=1}^{n} \alpha_i y_i \Phi(\mathbf{x}_i) + b$$

$$= \sum_{i=1}^{n} \alpha_i y_i \Phi(\mathbf{x}_i)\cdot\Phi(\mathbf{x}) + b$$

$$= \sum_{i=1}^{n} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$$

$$= \sum_{i=1}^{n} \alpha_i y_i \exp\left(-\frac{\|\mathbf{x}-\mathbf{x}_i\|^2}{2\sigma^2}\right) + b$$

Equivalent RBFNN



Discussion

- **Pros:**
  - Find the optimal separation hyperplane.
  - Can deal with very high dimensional data.
  - Usually work very well.
- **Cons:**
  - Require both positive and negative examples.
  - Need to select a good kernel function.
  - Require lots of memory and CPU time.
  - There are some numerical stability problems in solving the constrained QP.

Discussion

- **Steps to use SVMs**
  1. Choose a kernel function
     - RBF Kernels are mostly used
     - To choose proper parameters, use k-fold validation
  2. Choose a value for C
     - To choose a proper value, use k-fold validation
  3. Solve the quadratic programming problem (many software packages available)
  4. Construct the discriminant function from the support vectors